

BAKCHAR, Russian Federation

WMO: 293280

Lat: 57.0078N

Lon: 82.0603E

Elev: 361

StdP: 14.51

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -37.5	(c) -31.5	(d) -42.3	(e) 0.5	(f) -37.1	(g) -36.2	(h) 0.7	(i) -31.2	(j) 20.4	(k) 13.9	(l) 18.6	(m) 15.8	(n) 1.9	(o) 10	(p) 0.640

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.5	(c) 83.3	(d) 65.5	(e) 80.3	(f) 64.4	(g) 77.3	(h) 63.0	(i) 68.7	(j) 78.3	(k) 67.0	(l) 76.2	(m) 65.2	(n) 73.8	(o) 7.2	(p) 30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.4	95.2	73.1	63.5	89.0	71.3	61.7	83.4	69.5	33.2	78.4	31.8	76.3	30.4	73.7	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	16.3	14.2	-43.0	88.1	5.9	3.1	-47.2	90.4	-50.7	92.2	-54.0	94.0	-58.2	96.2
			-42.7	71.4	5.2	1.8	-46.4	72.8	-49.5	73.8	-52.4	74.9	-56.2	76.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	32.8	-3.0	4.0	19.8	35.8	49.4	60.6	64.0	59.0	47.7	35.0	15.9	3.2	(6)
(7)		DBStd	25.79	16.11	14.23	12.04	10.62	10.02	7.71	5.81	6.54	7.91	9.37	15.04	16.29	(7)
(8)		HDD50	7538	1644	1287	937	435	138	12	0	7	136	470	1022	1450	(8)
(9)		HDD65	11886	2109	1707	1402	875	491	169	89	206	521	930	1472	1915	(9)
(10)		CDD50	1253	0	0	0	10	120	330	435	286	67	5	0	0	(10)
(11)		CDD65	126	0	0	0	0	8	37	59	21	1	0	0	0	(11)
(12)		CDH74	1455	0	0	0	1	159	467	585	224	19	0	0	0	(12)
(13)		CDH80	285	0	0	0	0	36	97	114	37	1	0	0	0	(13)
(14)	Wind	WSAvg	6.6	6.7	7.2	7.9	7.5	7.2	5.6	4.9	5.1	5.8	6.7	7.2	7.2	(14)
(15)	Precipitation	PrecAvg	23.7	1.4	1.0	1.3	1.3	1.9	2.7	3.0	2.7	1.9	2.2	2.4	2.0	(15)
(16)		PrecMax	31.5	2.6	1.7	3.0	2.8	4.3	4.6	5.0	5.3	3.5	4.1	3.8	3.8	(16)
(17)		PrecMin	17.6	0.2	0.1	0.2	0.4	0.5	1.0	0.5	1.1	0.6	1.0	1.1	0.6	(17)
(18)		PrecStd	3.4	0.6	0.4	0.7	0.6	0.9	1.0	1.1	0.9	0.7	0.8	0.7	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.1	34.1	47.6	69.1	84.7	86.8	87.3	84.4	76.6	64.0	43.7	33.8	(19)
MCWB			29.5	32.2	37.6	51.7	59.7	65.4	68.8	67.7	61.4	50.1	41.0	32.4	(20)	
2%		DB	27.4	30.5	42.2	62.8	78.3	82.7	83.1	79.9	70.8	58.0	39.9	31.5	(21)	
		MCWB	26.0	28.5	34.7	48.8	57.5	64.5	67.1	65.1	57.6	47.8	37.2	30.5	(22)	
5%		DB	23.8	27.7	38.5	57.3	73.5	79.5	80.1	75.9	66.1	52.8	36.4	28.4	(23)	
		MCWB	22.6	25.9	33.1	45.8	55.6	63.8	65.5	63.4	55.5	45.5	34.3	27.2	(24)	
10%		DB	20.5	24.4	35.5	51.9	67.9	75.9	77.2	72.3	61.5	48.0	33.5	24.8	(25)	
		MCWB	19.4	22.8	31.6	42.7	52.7	62.3	64.2	61.7	52.7	43.0	32.2	23.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.5	32.3	38.7	53.8	62.5	69.8	71.9	70.1	62.5	52.1	41.6	32.8	(27)
MCDWB			30.6	34.0	46.1	64.8	77.6	80.3	82.7	79.3	71.9	59.0	43.0	33.5	(28)	
2%		WB	26.0	28.8	35.9	49.8	59.5	67.4	69.4	67.3	58.9	49.4	37.6	30.4	(29)	
		MCDWB	27.3	30.4	40.1	61.7	75.1	78.1	78.5	76.9	68.3	56.0	39.7	31.3	(30)	
5%		WB	22.7	26.1	33.9	46.5	57.0	65.5	67.7	64.9	56.4	46.1	34.5	27.1	(31)	
		MCDWB	23.7	27.5	37.8	56.6	70.2	75.5	76.5	72.6	64.4	52.3	36.0	28.2	(32)	
10%		WB	19.4	22.7	31.8	42.9	54.3	63.6	66.1	62.8	53.8	43.1	32.4	23.7	(33)	
		MCDWB	20.5	24.5	35.4	51.4	66.5	73.0	74.5	70.1	60.1	47.5	33.7	24.8	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.0	18.2	19.6	18.7	21.3	20.4	19.5	19.3	17.4	12.9	11.9	13.9	(35)
MCDBR			13.3	12.1	19.8	26.7	29.4	27.5	24.9	25.8	25.8	21.9	10.0	12.2	(36)	
MCWBR			13.1	11.2	14.2	15.1	13.0	12.2	10.9	12.8	13.8	13.2	9.1	11.9	(37)	
5% WB		MCDWB	13.2	11.4	15.7	25.5	26.0	23.2	21.2	21.6	22.8	19.6	9.8	11.9	(38)	
		MCWBR	13.0	11.0	12.1	15.1	13.5	11.2	10.4	11.8	13.1	12.9	9.2	11.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.284	0.318	0.368	0.398	0.402	0.406	0.442	0.408	0.367	0.346	0.312	0.286	(40)	
taud		2.014	1.996	1.933	2.094	2.243	2.308	2.196	2.282	2.351	2.260	2.085	1.962	(41)		
Ebn at Noon		174	220	239	256	265	265	251	252	246	213	161	132	(42)		
Edh at Noon		20	31	43	43	40	38	42	36	29	24	18	16	(43)		
(44)	All-Sky Solar Radiation	RadAvg	234	548	1011	1408	1611	1822	1653	1309	869	434	202	138	(44)	
(45)		RadStd	25	39	69	117	158	189	130	119	114	40	37	22	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46) N/A
(47)	Regional (7 neighbors)	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(47) N/A

Nomenclature: See separate page